

VLSI Design EE671 ASSIGNMENT-1

Note:

1. Plots and Graphs

- Ensure that all plots included in your report are **clearly legible and readable**.
- Each axis must be labelled with appropriate **values and units**.

2. Tools and PDK Requirement

- Install and use **IIC-OSIC-TOOLS** for all simulations.
- Use only the **Sky130A PDK** for this assignment.

3. Submission Format

- Submit your work as a **ZIP file** containing the following:
 - All code files used for the simulations.
 - A **Report in PDF format** including:
 - Code snippets.
 - Graphs/plots with results (**screenshots**).
 - Explanations and observations.

4. Simulation Parameters

- For all simulations, use a supply voltage of **VDD = 1.8 V**.

Introduction:

In this assignment you will design and simulate CMOS inverters using SkyWater 130A PDK.

You will use NGSpice (a spice circuit simulator) tool to simulate the circuit. NGSpice takes a spice netlist file as an input. A sample netlist is uploaded on moodle.

You need to create the netlist for the schematic shown in Fig. 1 including the MOSFETs and their dimensions (Width, W and Length, L). In addition to this, we want to include the MOSFET drain and source capacitance using spice area parameters "as", "ad" and perimeter parameter "pd". These will depend on the transistor geometry. Referring to the Fig. 2, the Width of the diffusion region is W while its length is 4λ . Since $\lambda = L_{min}/2$, $4\lambda = 2L_{min}$.

- The source/drain area parameters are: $as = ad = W \times 2 L_{min}$
- The source/drain perimeters are: $ps = pd = 2 \times (W + 2L_{min})$.

Notice that the values of area and perimeter of source and drain regions will have to be recalculated when you change the widths.

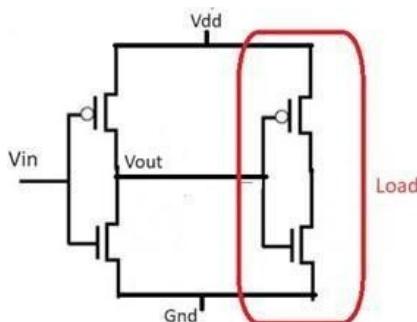


Figure 1: Inverter loaded with another inverter

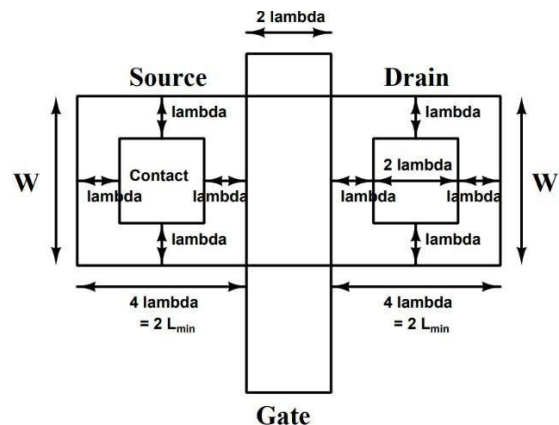


Figure 2: Illustration of diffusion area and perimeter

Q1) Design a minimum size CMOS inverter (minimum possible equal rise and fall time) with fixed NMOS and PMOS channel length of $0.15\mu\text{m}$ (this is the minimum channel length). The minimum channel width for NMOS is fixed to $0.42\mu\text{m}$ (restriction from the PDK) and the width of PMOS is to be obtained by design. The input should be a rail-to-rail square wave with rise/fall times of 20 ps, and the inverter will be loaded by another minimum size inverter (as shown in Fig. 1).

Simulate and report the following in the tabular fashion shown below:

Inverter Design Parameter	Value
PMOS Width (μm)	
PMOS Length (μm)	0.15
NMOS Width (μm)	0.42
NMOS Length (μm)	0.15

Inverter Dynamic Characteristic	Value
Rise time, t_r (ps)	
Fall time, t_f (ps)	
Propagation delay, t_p (ps)	

Once designed, **plot** (in the submission) the static transfer characteristics of this inverter by using a DC sweep on the input from 0 to VDD and report the following in the tabular fashion below:

Inverter Static Characteristic	Value
V_{IH} (V)	
V_{IL} (V)	
NM_H (V)	
NM_L (V)	
Switching Voltage, V_M (V)	

Q2) Let's call the inverter in Q1 as **INVX1** (strength-1 inverter). Design a strength-2 inverter (**INVX2**) and report all the tabular parameters above when the **INVX2** is loaded with **INVX1**. Comment on the results obtained.